Akoizo Original Practice - Form 2 - Answer key

Division B | Solar System | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A planet moves from 1 AU to 2 AU. By what factor does the starlight flux change?

A. It doubles

B. It becomes 1/8 as large

C. It becomes 1/4 as large

D. It becomes 1/2 as large

Correct: C

2. (2 points) Which moon has its own internally generated magnetic field?

Unit: Ocean worlds: Europa, Ganymede, Enceladus, and Titan.
Worked-case data: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet's field. What does this suggest about the moon's interior?
Separate investigation observations:
Atmosphere: Thick nitrogen, 1.5 times Earth's surface pressure.
Surface temperature: About -180 °C.
Radar: Smooth, dark lakes near the poles.
Probe images: Rounded pebbles on the landing site.

A. Titan

B. Enceladus

C. Ganymede

D. Europa

Correct: C

3. (2 points) Why are small, hot worlds poor at keeping light gases?

Unit: What makes a world habitable?.
Worked-case data: Europa orbits Jupiter far outside the Sun’s habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.
Separate investigation observations:
Location: The moon orbits Saturn, far outside the habitable zone.
Plume: Jets of water vapor and ice grains erupt from cracks.
Plume chemistry: Salts, silica grains, organic molecules, and hydrogen gas.
Orbit: An elliptical orbit kept by resonance with another moon.

A. Gas molecules move fast compared with a low escape velocity

B. Gravity is too strong

C. Light gases freeze out

D. The Sun adds gas

Correct: A

4. (2 points) A high D/H ratio in Venus’s atmosphere suggests:

Unit: Venus and Mars: rocky neighbors.
Worked-case data: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?
Separate investigation observations:
Orbital image: A fan of layered sediment where a valley enters a crater.
Spectra: Clay minerals and carbonates detected in the fan.
Rover data: Fine-grained mudstone layers on the crater floor.

A. Venus lost a lot of water over time

B. Venus has oceans today

C. Comets just delivered water

D. The atmosphere is mostly hydrogen

Correct: A

5. (2 points) Which telescope used transmission spectroscopy to study TRAPPIST-1 planets?

A. JWST

B. Kepler

C. Rosetta

D. MAVEN

Correct: A

6. (2 points) Tidal force depends on distance as:

Unit: Gravity, orbits, and tides.
Worked-case data: Planet A has 4 times Earth's mass and 2 times Earth's radius. Compare its surface gravity and escape velocity with Earth's.
Separate investigation observations:
Orbit: Moon B orbits its planet in 3.6 days on a slightly elliptical path.
Rotation: It always shows the same face to the planet.
Neighbors: Its period is exactly twice that of an inner moon.
Surface: Young, crack-covered ice with few craters.

A. 1/d^2

B. d^2

C. 1/d^3

D. 1/d

Correct: C

7. (2 points) An organism at a dark hydrothermal vent that uses hydrogen sulfide for energy is a:

Unit: Small bodies and the building blocks of life.
Worked-case data: Earth's oceans have a D/H ratio of about 1.6 x 10□^4. Comet 67P's water has about 5.3 x 10□^4, and carbon-rich asteroids have values close to Earth's. Which source better matches Earth's water?
Separate investigation observations:
Sample minerals: Clay minerals and sodium carbonate salts.
Organics: Amino acids and nucleobases detected in the lab.
Isotopes: D/H ratio close to Earth's oceans.

A. Tardigrade

B. Heterotroph that eats plants

C. Chemolithoautotroph

D. Photoautotroph

Correct: C

8. (4 points) Why is Earth's actual average temperature higher than its equilibrium temperature?

answer: Greenhouse gases absorb outgoing infrared radiation and re-emit some back toward the surface, warming it above the 255 K equilibrium value. (4 points).

9. (4 points) Why is Europa considered a strong candidate for habitability?

Unit: Ocean worlds: Europa, Ganymede, Enceladus, and Titan.
Worked-case data: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet's field. What does this suggest about the moon's interior?
Separate investigation observations:
Atmosphere: Thick nitrogen, 1.5 times Earth's surface pressure.
Surface temperature: About -180 °C.
Radar: Smooth, dark lakes near the poles.
Probe images: Rounded pebbles on the landing site.

answer: It likely has a large salty ocean in contact with a rocky seafloor, kept liquid by tidal heating, with possible chemical energy from water-rock reactions. (4 points).

10. (4 points) Describe how orbital resonance keeps tidal heating going.

Unit: What makes a world habitable?.
Worked-case data: Europa orbits Jupiter far outside the Sun's habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.
Separate investigation observations:
Location: The moon orbits Saturn, far outside the habitable zone.
Plume: Jets of water vapor and ice grains erupt from cracks.
Plume chemistry: Salts, silica grains, organic molecules, and hydrogen gas.
Orbit: An elliptical orbit kept by resonance with another moon.

answer: Regular gravitational tugs from the other moons keep each orbit elliptical, so the moon keeps flexing as its distance from the planet changes. (4 points).

11. (4 points) How did losing its magnetic field affect Mars?

Unit: Venus and Mars: rocky neighbors.
Worked-case data: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?
Separate investigation observations:
Orbital image: A fan of layered sediment where a valley enters a crater.
Spectra: Clay minerals and carbonates detected in the fan.
Rover data: Fine-grained mudstone layers on the crater floor.

answer: Without a global magnetic field, the solar wind could strip gas from the upper atmosphere, thinning it over billions of years. (4 points).

12. (4 points) Why can microlensing not be used to follow up a planet?

Unit: Exoplanets: detection and the named systems.
Worked-case data: An Earth-sized planet transits two stars: a Sun-like star and a red dwarf with one-tenth the Sun's radius. Compare the transit depths.
Separate investigation observations:
Light curve: Seven different periodic dips in brightness.
Star: An ultracool red dwarf, about 40 light-years away.
Sizes: All planets are roughly Earth-sized.
JWST: The two innermost planets show no thick atmosphere.

answer: The alignment of the foreground and background stars happens only once, so the event cannot be repeated. (4 points).

13. (4 points) Why do planets in a red dwarf's habitable zone have short orbital periods?

answer: The habitable zone is very close to a dim star, and Kepler's third law gives short periods for small orbits. (4 points).

14. (4 points) Explain why extremophiles matter for habitability studies.

Unit: Small bodies and the building blocks of life.
Worked-case data: Earth's oceans have a D/H ratio of about 1.6 x 10□^4. Comet 67P's water has about 5.3 x 10□^4, and carbon-rich asteroids have values close to Earth's. Which source better matches Earth's water?
Separate investigation observations:
Sample minerals: Clay minerals and sodium carbonate salts.
Organics: Amino acids and nucleobases detected in the lab.
Isotopes: D/H ratio close to Earth's oceans.

answer: They show life can survive without sunlight and in cold, salty, or high-pressure conditions, widening the places where life might exist. (4 points).

15. (4 points) A spectral line appears at a shorter wavelength than expected. What does this mean?

Unit: Light, radiation, and temperature.
Worked-case data: Planet X orbits a Sun-like star at 4 AU with albedo 0.3, the same as Earth. Estimate its equilibrium temperature compared with Earth's 255 K.
Separate investigation observations:
Distance: A planet orbits a Sun-like star at 0.72 AU.
Albedo: Very high: about 0.75, from bright clouds.
Predicted Teq: About 230 K, below freezing.
Measured surface: About 735 K.

answer: The source is moving toward the observer: a blueshift. (4 points).

16. (4 points) What is unique about JUICE's planned orbit?

Unit: Ocean worlds: Europa, Ganymede, Enceladus, and Titan.
Worked-case data: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet's field. What does this suggest about the moon's interior?
Separate investigation observations:
Atmosphere: Thick nitrogen, 1.5 times Earth's surface pressure.
Surface temperature: About -180 °C.
Radar: Smooth, dark lakes near the poles.
Probe images: Rounded pebbles on the landing site.

answer: It will be the first spacecraft to orbit a moon of another planet, Ganymede. (4 points).

17. (6 points) Solve this case and explain your reasoning: Europa orbits Jupiter far outside the Sun's habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.

reasoning: Sunlight there is too weak to keep surface water liquid.
Europa's elliptical orbit, maintained by resonance with Io and Ganymede, causes tidal flexing and internal heating. An ice shell insulates the ocean and provides pressure, keeping water liquid below. (3 points).

conclusion: Tidal heating plus an insulating ice shell can keep an ocean liquid without sunlight, so habitability is not limited to the habitable zone. (3 points).

18. (6 points) Solve this case and explain your reasoning: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?

reasoning: A channel ending in a fan inside a crater suggests a river delta.
Clay minerals form when water alters rock over time.
Layering indicates sediments were deposited in standing water. (3 points).

conclusion: The crater likely held a long-lived lake fed by a river, a potentially habitable environment worth sampling. (3 points).